

Predictive coding cap experiments

Learning corrections on a frozen transformer

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Retrieval, learning signals, and the consequences of extreme errors.

Can a small cap learn without disrupting the rest?

Teach a frozen language model new answers while preserving its other predictions.

Active inference

The longer-term goal: update beliefs and choose informative actions.

Predictive coding

Test whether iterative error inference provides useful learning signals.

Heavy-tailed distributions

Ask whether rare, large losses change the story told by averages.

We tested components of this programme. The full active-inference agent remains open.

What a cap is

GPT-2 small: 12 blocks, 124M parameters. Base weights stay fixed during the edit experiments.

Frozen transformer

Prompt → blocks 1-12 → next-token probabilities



Cap and memory

Read activations → retrieve a correction or abstain → add bounded write vectors

Read/write sites: after blocks 4, 8 and 12.

One local GPU; transfer to larger models is untested.

No correction reproduces the base. Frozen weights can still produce changed behavior.

Which reader finds the correction?

Condition	Retrieval rule
Stable v0	Fixed activation-distance thresholds select stored corrections.
Learned reader	A trained reader selects a relevant correction or chooses no correction.
Random-reader control	Untrained random geometry, with its specified gate configuration.

The main learned reader was trained by backpropagation before these edit streams.

Reusable reader weights stay fixed; the correction memory continues to learn.

Teach, test, then retest after more learning

Teach a fact → test it now → teach more facts → retest earlier facts and paraphrases.

Main-study dataset	Edits per stream
zsRE	1,000
CounterFact	1,000
MQuAKE	300

Usefulness

Does the original question—and a held-out rewording—still produce the taught answer?

Preservation

Do unrelated questions retain their answers? How do predictions change on ordinary text?

Three fact samples, each tested in five orders. Our MQuAKE assay uses single-fact edits.

Original-prompt retention and paraphrase retention answer different questions.

Remembering a prompt is not enough

Dataset	Stable v0: original prompt	Stable v0: paraphrase	Learned reader: paraphrase
zsRE · 1,000 edits	66.68%	18.56%	96.03%
CounterFact · 1,000 edits	100%	0%	67.80%
MQuAKE · 300 edits	100%	0%	71.56%

Means across orders within each of three fact samples, then across samples.

Four-sample extension: learned versus random reader gained 44.1 points on zsRE and 56.2 on CounterFact in paraphrase retention.

Learned retrieval greatly improved access to corrections under new wording.

Better retrieval did not meet every preservation goal

45 / 45 cells

The learned reader exceeded the registered preservation limit: mean $KL \leq 0.001$ nats per token on ordinary text.

CounterFact

The comparison with stable v0 remained formally inconclusive because its locality requirement failed.

MQuAKE

The shorter 300-edit run remains descriptive.

KL measures change in the full next-token probability distribution.

A large paraphrase gain is not a complete preservation success.

Where predictive coding enters

Teaching a fact needs a direction for changing its stored correction.

Adjoint credit

Differentiate answer loss through the frozen transformer. Use the negative normalized gradient.

Error-based PC credit

Relax temporary error variables against prediction loss and an error penalty. Use the settled site errors to direct the correction.

Our digital solver uses automatic differentiation through the graph.

We separately tested PC credit for acquiring facts and for training the reader/controller.

PC is a tested learning mechanism here; autonomous action selection is still future work.

Changing acquisition credit gave little retention gain

Dataset	Credit	Paraphrase retention	Whole-process time	Max. loss increase vs. base
zsRE	Adjoint	98.3%	316 s	9.26 nats
zsRE	Error inference	98.3%	426 s	12.27 nats
CounterFact	Adjoint	80.7%	297 s	11.74 nats
CounterFact	Error inference	80.3%	359 s	12.23 nats

Mean harm moved in different directions across the two datasets. Small endpoint differences do not prove equivalence.

This changes how facts are acquired; it does not retrain the reader.

PC training reduced paraphrase retention in 5 of 6 pairs

Dataset and seed	BP retention	ePC retention
zsRE · 0	97.3%	94.7%
zsRE · 1	99.0%	97.0%
zsRE · 2	97.7%	98.0%
CounterFact · 0	81.0%	56.8%
CounterFact · 1	80.0%	77.8%
CounterFact · 2	83.5%	66.0%

96-102× training time

ePC: 23-25 hours per seed. BP: about 15 minutes.

Controlled comparison

300 updates; 3.35M reader/controller parameters. Every evaluation uses adjoint acquisition.

Some harm measures improved, but PC was much more costly in this configuration.

More settling improved original-prompt retention

Credit / settling steps	Original prompt	Paraphrase	Learning time
Adjoint reference	51.5%	13.1%	4.6 min
Error inference · 1	51.5%	13.1%	5.7 min
Error inference · 8	53.4%	13.1%	10.2 min
Error inference · 32	56.5%	12.2%	25.4 min

Mean ordinary-text loss increase rose from 0.00145 to 0.00221 nats between 1 and 32 steps.

124M-parameter base distilled over 50M training tokens. Three exposed fact samples, one order; a separate comparison from the main study.

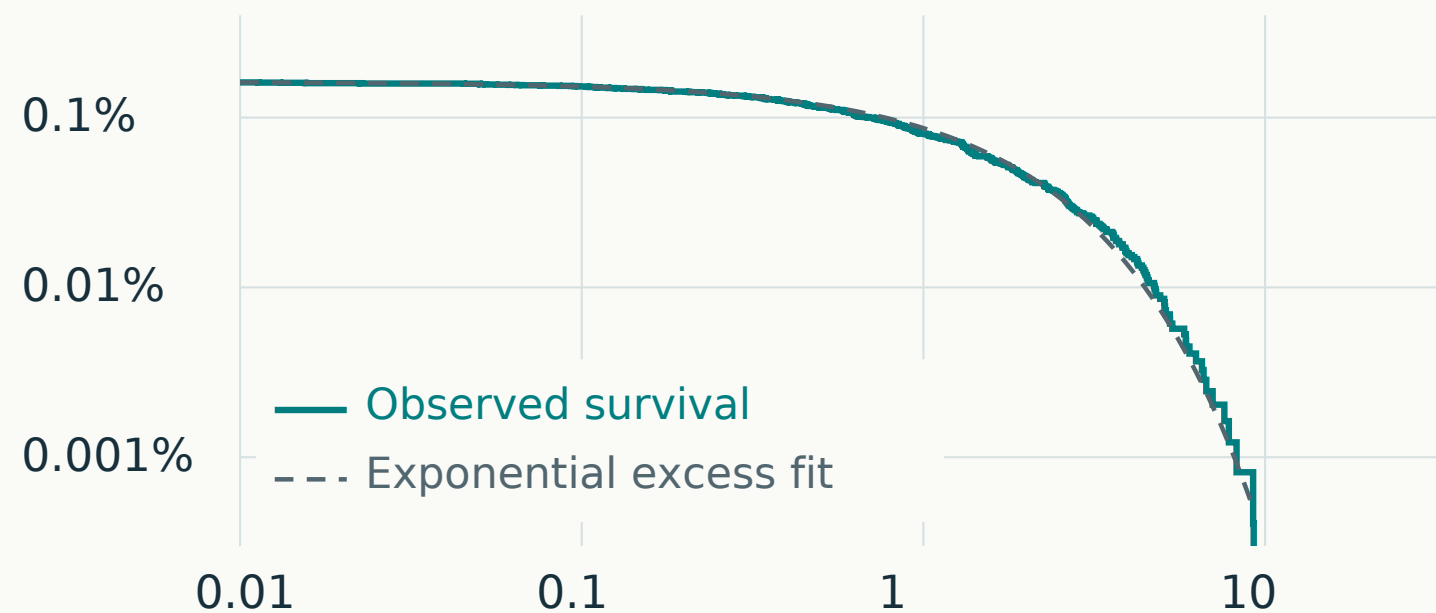
The original-prompt gain did not extend to paraphrases, and cost more time and harm.

Extreme losses have different observed tails

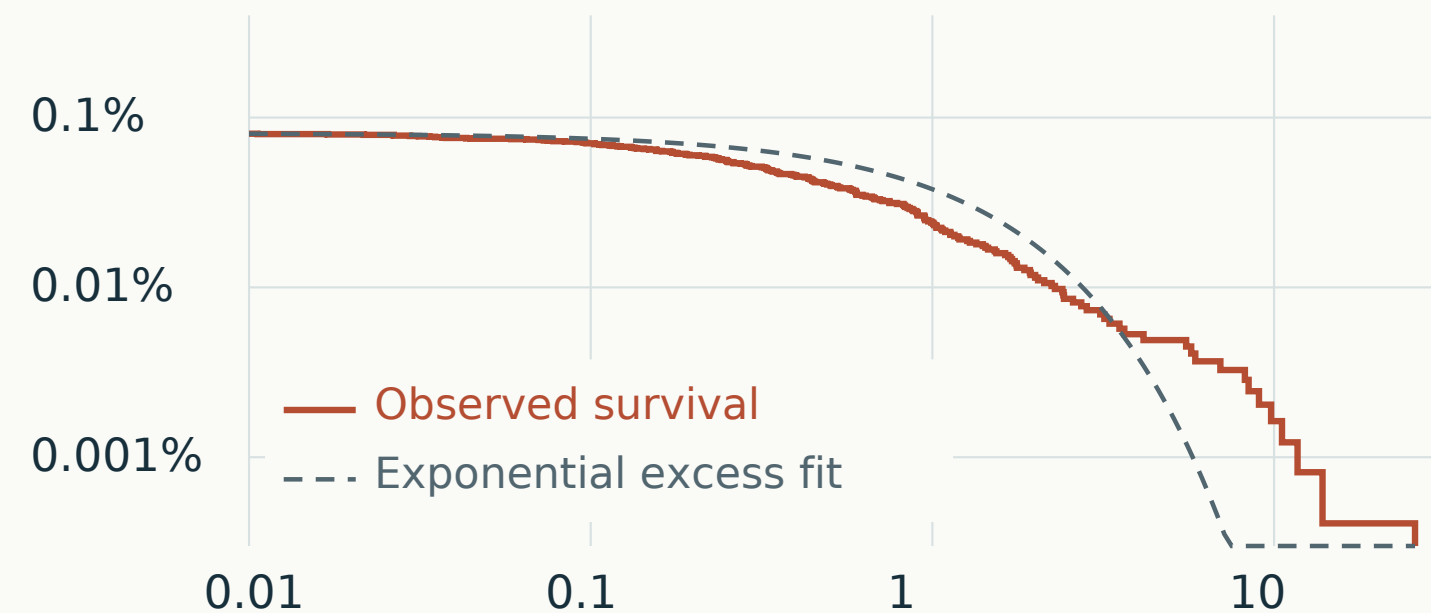
Δ = loss with cap – loss without cap, at the same prefix. Positive Δ means worse prediction.

Positions above threshold (log scale)

zsRE · learned reader



zsRE · stable v0



Loss increase Δ (nats; logarithmic scale)

1 nat = a probability decrease by a factor of about 2.72. Fits use excesses above 0.01 nat.

Stable v0 has the more pronounced tail here. A power-law family is not established.

A simple mixture bounds token loss increase

$$p_{\text{mix}} = \rho p_{\text{base}} + (1 - \rho) p_{\text{cap}}, \text{ where } \rho = e^{-1} \approx 0.37$$

At the same prefix: $p_{\text{mix}} \geq \rho p_{\text{base}}$, so loss increase over the base is at most 1 nat.

Conditional severity	Original cap	Mixture
zsRE	1.619 nats	0.542 nats
CounterFact	1.925 nats	0.581 nats

Severity averages positions with $\Delta > 0.01$ nat. Ten exposed streams; 300 edits.

Paraphrase retention: unchanged on zsRE; down 0.7–1.2 points on CounterFact. CounterFact still missed the mean-KL limit.

The guarantee bounds per-token loss increase; unchanged greedy answers are not guaranteed.

Toward active inference in the extremes

Established components

We measured retrieval, PC learning trade-offs, and concentrated prediction losses.

Open scientific step

Can an active-inference policy choose which facts to audit or update more effectively under the same budget?

Autonomous action selection and a tested probabilistic Markov blanket remain open.

We welcome collaboration on a precise coupled-entropy model and a discriminating experiment. The full coupled theory remains untested here.

Evaluate usefulness, the learning mechanism, and extreme consequences together.

Questions and discussion

earthland.ai/ccs/

Canonical Google deck and presentation links

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SingularityNET · EarthLand · BGI Labs · TrueAGI

Predictive-coding implementation: FabricPC

Code: github.com/pderp/pc_cap

Materials: github.com/pderp/cap-assets

Agentic assistance: Capex · Capstan · Derp Peoples



Which mechanism would you test next—and which failure would matter most?